PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 3 · Version B

20 questions · 90 minutes · original practice exam, not an official exam

Name _____ NetID _____ Section _____

Closed book. Calculators allowed. Use the official PHY 2111 formula sheet. Take $g = 9.81\ m/s^2$.
MC5 questions have five choices (6 points), MC3 three choices (3 points), MC2 two choices (2 points).
Mark one answer per question. Answers and worked solutions are in the separate solutions file.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A door is 1.0 m wide. You push with a 40 N force at a point 0.50 m from the hinges, at 45° to the door.

1) What torque do you exert about the hinges? **MC5**

- **a.** 14.1 N·m
- **b.** 20 N·m
- **c.** 40 N·m
- **d.** 10 N·m
- **e.** 28.3 N·m

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

An Atwood machine has masses of 3.0 kg and 5.0 kg on either side of a pulley that is a uniform solid disk (M = 4.0 kg, R = 0.15 m). The rope does not slip and the axle is frictionless.

2) What is the magnitude of the acceleration of the masses? **MC5**

- **a.** 2.45 m/s^2
- **b.** 0.981 m/s^2
- **c.** 1.96 m/s^2
- **d.** 1.64 m/s^2
- **e.** 3.92 m/s^2

3) What is the tension in the rope on the 3.0 kg side? **MC5**

- **a.** 36.8 N
- **b.** 49.1 N
- **c.** 29.4 N
- **d.** 35.3 N
- **e.** 39.2 N

4) How do the tensions on the two sides compare? **MC3**

a. They are equal.
b. The tension on the 3.0 kg side is larger.
c. The tension on the 5.0 kg side is larger.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A thin hoop and a solid cylinder, with the same mass and radius, are released together from rest at the top of the same incline and roll without slipping. The top is 2.0 m above the bottom.

5) Which reaches the bottom first? **MC3**

a. The solid cylinder
b. They tie
c. The hoop

6) What is the speed of the hoop at the bottom? **MC5**

a. 5.11 m/s
b. 3.13 m/s
c. 4.43 m/s
d. 5.29 m/s
e. 6.26 m/s

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A grinding wheel is a uniform disk of mass 5.0 kg and radius 0.20 m, spinning at 300 rad/s. A tool is pressed against its rim with a normal force of 40 N, and the coefficient of kinetic friction is 0.60.

7) What is the magnitude of the angular acceleration of the wheel? **MC5**

a. 48 rad/s²
b. 96 rad/s²
c. 24 rad/s²
d. 240 rad/s²
e. 4.8 rad/s²

8) How long does the wheel take to stop? **MC5**

a. 0.16 s
b. 3.12 s
c. 12.5 s
d. 62.5 s
e. 6.25 s

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A uniform horizontal beam 4.0 m long with a mass of 20 kg is hinged to a wall at one end. A cable attached to the other end makes a 30° angle with the beam. A 50 kg sign hangs from the end of the beam.

9) What is the tension in the cable? MC5

a. 1180 N
b. 589 N
c. 1360 N
d. 687 N
e. 2350 N

10) What is the horizontal force the hinge exerts on the beam? MC5

a. 1180 N
b. 589 N
c. 1020 N
d. 0 N
e. 687 N

11) If the cable were attached to the wall higher up, so it made a larger angle with the beam, the tension would MC3

a. increase.
b. stay the same.
c. decrease.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

Disk A ($I = 0.50\ \text{kg·m}^2$) spins at 30 rad/s. Disk B ($I = 1.0\ \text{kg·m}^2$), initially not spinning, is dropped onto it along the same axle, and the disks stick together.

12) What is the final angular speed? MC5

a. 15 rad/s
b. 20 rad/s
c. 10 rad/s
d. 30 rad/s
e. 7.5 rad/s

13) What fraction of the initial kinetic energy is lost? MC5

a. 66.7%
b. 100%
c. 33.3%
d. 50.0%
e. 0%

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 2.0 kg ball moves at a constant 5.0 m/s in the +x direction along the line y = 3.0 m.

14) What is the magnitude of the ball's angular momentum about the origin? **MC5**

- **a.** 50 kg·m²/s
- **b.** 10 kg·m²/s
- **c.** 15 kg·m²/s
- **d.** 30 kg·m²/s
- **e.** 0 kg·m²/s

15) As the ball moves along the line, its angular momentum about the origin **MC3**

- **a.** decreases.
- **b.** stays constant.
- **c.** increases.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.50 kg block on a spring moves according to x(t) = (0.12 m) cos(8.0t), with t in seconds.

16) What is the spring constant? **MC5**

- **a.** 128 N/m
- **b.** 16 N/m
- **c.** 32 N/m
- **d.** 0.25 N/m
- **e.** 4 N/m

17) What is the total mechanical energy of the oscillation? **MC5**

- **a.** 0.0288 J
- **b.** 1.92 J
- **c.** 0.461 J
- **d.** 0.115 J
- **e.** 0.23 J

18) If the mass were four times larger (same spring), the period would **MC3**

- **a.** double.
- **b.** quadruple.
- **c.** be cut in half.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A 0.30 kg mass hangs at rest from a vertical spring, which it stretches by 6.0 cm.

19) If the mass is pulled down slightly and released, what is the period of oscillation? **MC5**

a. 0.491 s
b. 0.347 s
c. 0.0782 s
d. 0.983 s
e. 2.04 s

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

You want to build a simple pendulum with a period of 1.5 s on Earth.

20) How long should the pendulum be? **MC5**

a. 0.559 m
b. 3.51 m
c. 2.34 m
d. 0.0891 m
e. 0.373 m
